

CE 310

Multiple Linear Regression

Week 10 · Session 1 · Adding Predictors: Occupancy, Season, and Interaction Effects

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Two Memos. One Dataset. Opposite Recommendations.

90 seconds · which one do you sign?

A state DOT is deciding whether to keep **Contractor A** on its prequalified list. Five years of condition data, 26 rehabilitated pavement sections:

Contractor	Sections	Mean PCI lost over 5 years
A	12	22.0 points
B	14	15.0 points

- **Engineer 1** — "A's sections lost 7 more points. Remove A from the list."
- **Engineer 2** — "Keep A."

Same 26 rows. The next award is worth \$14M. **Which memo goes out over your signature?**

✓ Both Read the Data Right. One Held Traffic Constant.

The raw gap is a truck-loading gap — and it hides the opposite contractor effect

The file has a third column. A's sections carry **8.4M ESALs** of truck loading; B's carry **3.1M**.

Fit both predictors at once: $\text{PCI lost} = 8.8 + 2.0 (\text{ESALs}) - 3.6 (\text{Contractor A})$

Source of the 7.0-point gap	Points
Traffic — A carries 5.3M more ESALs, at 2.0 points per million	+10.6
Contractor — at equal traffic loading	−3.6
Observed raw difference	+7.0

💡 At the same traffic loading, A's sections lose **3.6 points less**. The raw comparison did not overstate the gap — it reversed its sign.

Week 10 Roadmap

Three conceptual moves today — each unlocks a new engineering capability:

Move	Core idea	Engineering payoff
One → many predictors	Hold other variables constant	Isolate the true occupancy premium from temperature
Continuous → categorical	Dummy variables & collinearity	Know when adding a variable hurts more than it helps
Additive → interactive	Slopes that change by group	Size equipment for peak occupied conditions correctly

Data: same 8,760-hour Tucson DOE dataset — adding predictors one at a time.

Goal: interaction model at $R^2 \approx 0.795$, up from 0.601 in Week 9.

Where Week 9 Left Off

Week 9 fitted a single-predictor model: $\hat{y} = 1.1585 \cdot T + (-60.68)$, where T is outdoor temperature in °F.

Result: $R^2 = 0.601$ — outdoor temperature explains **60.1%** of the variance in cooling energy.

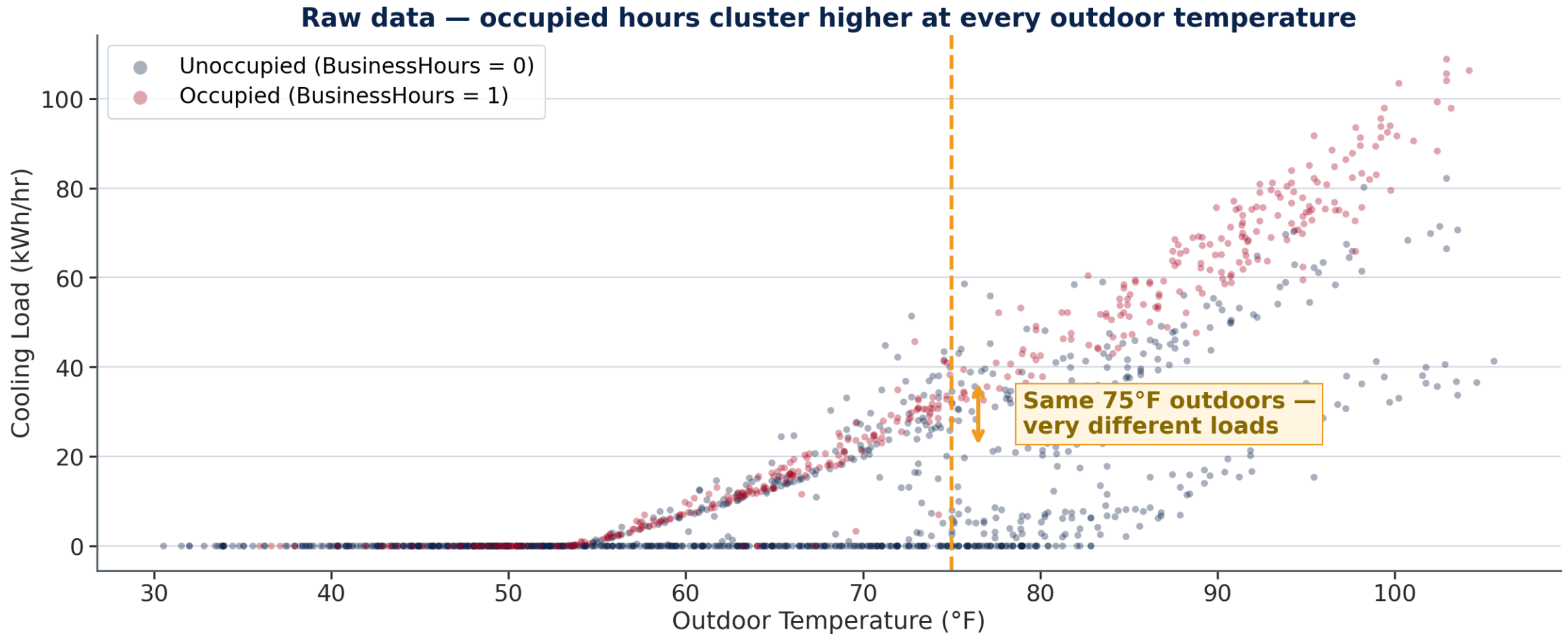
The question for today: **what explains the other 39.9%?**

Consider two hours with the same outdoor temperature — say, 75°F:

Hour	Outdoor Temp	Building	Actual Cooling
Monday 2 pm	75°F	200 people + all computers on	High
Sunday 2 am	75°F	Empty, setback mode	Low

Same temperature. Very different cooling load. Outdoor temperature alone cannot tell these apart.

Two Clouds in the Raw Data



The Missing Predictor: Occupancy

When we split the data by occupancy status:

Group	Mean Outdoor Temp	Mean Cooling (kWh/hr)
Occupied hours	~73°F	38.9 kWh/hr
Unoccupied hours	~64°F	10.6 kWh/hr
Raw difference		28.3 kWh/hr

Occupied hours are both **hotter outside** and **heavier on cooling** — two effects mixed together.

The 28.3 kWh/hr raw difference is not just occupancy — it includes the fact that business hours coincide with the hottest part of the day. Multiple regression separates these two effects.

The Multiple Regression Equation

Add a second predictor alongside outdoor temperature:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

For this dataset:

Symbol	Meaning	Value (Model 2)
x_1	Outdoor temperature (°F)	—
x_2	BusinessHours (1 = occupied, 0 = not)	—
β_1	kWh/hr per 1°F rise, holding occupancy constant	1.026 kWh/°F
β_2	Extra kWh/hr occupied, holding temperature constant	17.9 kWh/hr
β_0	Predicted cooling at 0°F, unoccupied	−55.5 kWh

Fitting MLR in Python

`scipy.stats.linregress` handles one predictor. For multiple predictors, use `statsmodels`:

The model is written the way you would say it aloud — **cooling as a function of outdoor temperature and business hours** — and fitting it returns a summary table with each coefficient, its standard error, and its p -value.

The tutorial covers the syntax and how to read the summary.

Output section	What to read
<code>coef</code> column	$\hat{\beta}$ estimates — one row per predictor
<code>P> t </code> column	p -value: is this coefficient distinguishable from zero?
<code>R-squared</code> / <code>Adj. R-squared</code>	Overall model fit

p -value rule of thumb: $p < 0.05$ means the predictor is real, not noise. Week 11 derives this from first principles.

What "Holding Constant" Means

The phrase *holding the other predictor constant* is the core idea of multiple regression.

Example: $\hat{y} = -56.87 + 1.026 \cdot T + 17.9 \cdot BH$

Prediction at 75°F, occupied (BH = 1):

$$\begin{aligned}\hat{y} &= -56.87 + 1.026(75) + 17.9(1) \\ &= -56.87 + 76.95 + 17.9 = 38.0 \text{ kWh/hr}\end{aligned}$$

Prediction at 75°F, unoccupied (BH = 0):

$$\begin{aligned}\hat{y} &= -56.87 + 1.026(75) + 17.9(0) \\ &= -56.87 + 76.95 + 0 = 20.1 \text{ kWh/hr}\end{aligned}$$

The **17.9 kWh/hr difference** is the occupancy premium — estimated at the **same outdoor temperature**.

This is what Excel LINEST and Week 9's `scipy.stats.linregress` cannot do: they fit one predictor at a time.

Confounding

Raw mean difference: occupied (38.9) – unoccupied (10.6) = **28.3 kWh/hr**

Regression BH coefficient: **17.9 kWh/hr** — where did the other 10.4 kWh go?

Occupied hours coincide with the hottest part of the day:

- Occupied: daytime, weekdays → outdoor temp averages ~73°F
- Unoccupied: nights, weekends → outdoor temp averages ~64°F
- The raw 28.3 kWh gap mixes two effects: occupancy AND the warmer daytime temperatures

Regression holds outdoor temperature constant and isolates the occupancy premium alone. The **17.9 kWh/hr** coefficient is a fairer estimate of what occupancy itself adds.

Part 2: Categorical Predictors & Collinearity

When variables are labels — and when adding one makes things worse

Reading the Output — Coefficients

`statsmodels` produces a detailed output. Here are the coefficient-level rows you need.
(**BH** = BusinessHours indicator: 1 = occupied, 0 = unoccupied)

Row/column	What it tells you	Example (Model 2)
<code>coef</code> column	Estimated slope for each predictor	BH coef = 17.9 kWh/hr
<code>std err</code> column	Uncertainty in the estimate	BH $\approx$ 0.34
<code>t</code> column	Coefficient divided by its std err	BH $\approx$ 52.7
<code>P> t </code> column	p-value: is this coefficient real or noise?	BH $\approx$ 0.000

Rule of thumb: if `P>|t|` < 0.05, the coefficient is statistically significant — the predictor is earning its place in the model.

Reading the Output – Model Fit Statistics

These rows summarise how well the whole model fits.

Row/column	What it tells you	Example (Model 2)
R-squared	Fraction of variance explained	0.697
Adj. R-squared	R ² penalized for extra predictors	0.697
F-statistic	Joint test: are any predictors useful?	large → model is significant

R² vs. Adjusted R²

R² always increases when you add a predictor, even a useless one.

Adjusted R² penalizes for extra predictors. If a new predictor adds less than its share of variance, Adj. R² stays flat or decreases slightly.

Always compare models using **Adjusted R²** when they have different numbers of predictors.

For this dataset ($n = 8760$), the penalty per predictor is tiny — R² and Adj. R² differ by less than 0.001 even in the largest models. The distinction matters more in small datasets or models with many predictors relative to observations.

Quick Check — Does Season Help?

60 seconds · pick A or B

Model 1 (Week 9) used outdoor temperature alone: $R^2 = 0.601$.

Model 2 added BusinessHours: $R^2 = 0.697$.

Now Season is added (Summer / Spring / Fall / Winter) alongside Outdoor_Temp_F.

What do you predict happens to R^2 ?

- **A)** R^2 jumps by another 9+ percentage points — similar gain to BusinessHours
- **B)** R^2 barely moves, staying close to 0.601 — Season adds almost nothing new

✓ Quick Check — Answer: B, Season Barely Helps

R^2 rises from 0.601 to only ~0.611

Model	Predictors	R^2	Gain
Model 1	Outdoor_Temp_F	0.601	—
+ BusinessHours	Outdoor + BH	0.697	+0.096
+ Season	Outdoor + Season	~0.611	+0.010

Why? In Tucson, "Summer" means "outdoor temperature is consistently high." Outdoor temperature already captures this — Season provides almost no new information. This is **collinearity**: two predictors that tell the same story.

💡 Adding a collinear predictor inflates the model, adds complexity, and makes coefficients unstable — without improving prediction.

Collinearity — When Two Predictors Tell the Same Story

Collinearity means two predictors contain overlapping information. Adding a collinear predictor barely changes R^2 .

Season and Outdoor_Temp_F are collinear:

Season	Mean Outdoor Temp (°F)
Summer	~90°F
Spring	~72°F
Fall	~62°F
Winter	~49°F

Collinear predictors also inflate standard errors — it becomes harder to tell which predictor is doing the work.

Detecting Collinearity: VIF

The **Variance Inflation Factor** quantifies how much each coefficient's uncertainty grows because of collinearity.

The **variance inflation factor** is computed for each predictor in turn. VIF above roughly 5 says that predictor is largely explained by the others, and its coefficient should not be read on its own.

The tutorial covers how to compute it.

Feature	VIF	Verdict
Outdoor_Temp_F	3.2	✓ OK — manageable correlation with Season
BH (business hours)	1.4	✓ OK — nearly independent of temperature
Season dummies	4.8–6.1	⚠ Concern — Season and Temp share information

Rule of thumb: $VIF > 5$ = concern · $VIF > 10$ = problem.

Dummy Variables — How a Category Becomes a Number

Regression can only work with numbers. To use a yes/no variable as a predictor, convert it:

A text column of "Yes"/"No" becomes a **0/1 indicator** — 1 when occupied, 0 when not. That single column is what lets a categorical fact enter a numeric model.

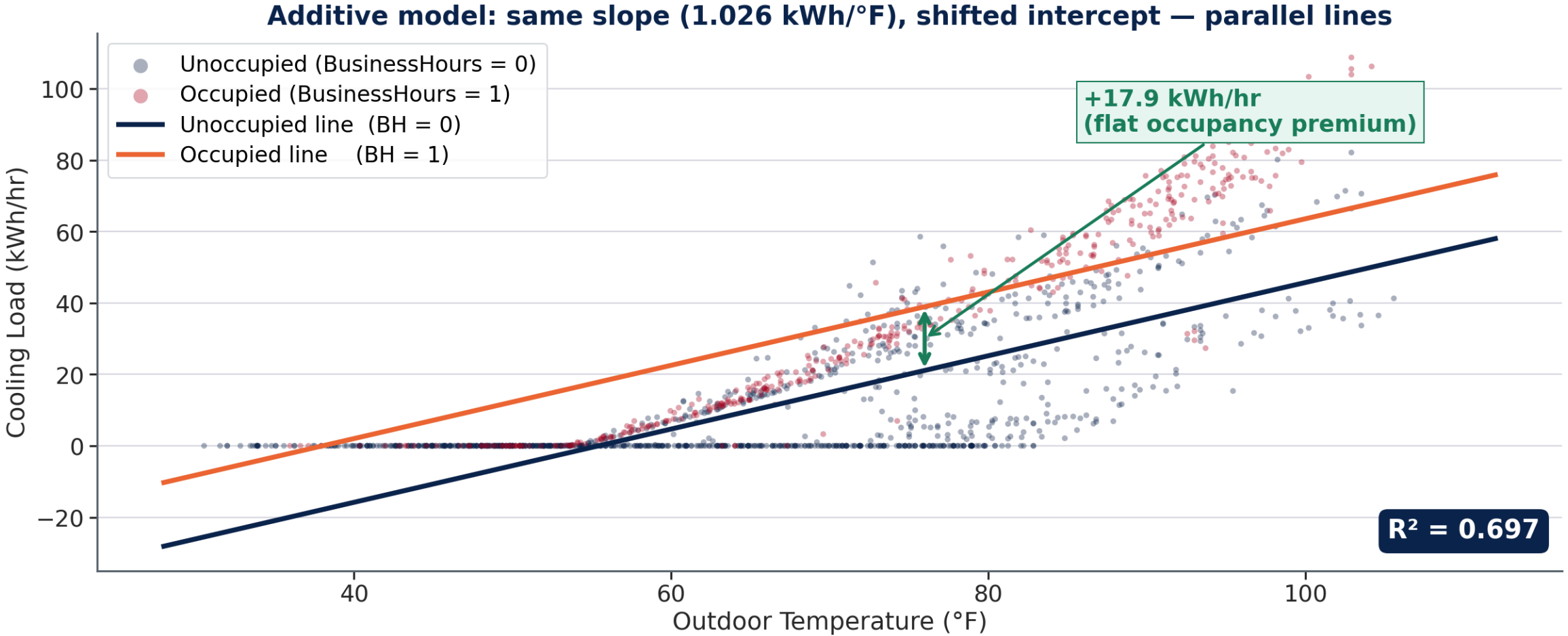
A **dummy variable** is a 0/1 switch used as a *predictor* in linear regression. (In logistic regression a binary variable is the *outcome* — different method, different role.)

In the regression equation:

- When BH = 0: $\hat{y} = \beta_0 + \beta_1 T$ (unoccupied line)
- When BH = 1: $\hat{y} = (\beta_0 + \beta_2) + \beta_1 T$ (occupied line — shifted up by β_2)

Both lines have the **same slope** (β_1). The only difference is the intercept. This is the **additive model** assumption — occupancy shifts the whole line up, but the temperature sensitivity is the same occupied or not.

Additive Model – Two Parallel Lines



Additive Model — What the Parallel Lines Claim

The model assigns the **same slope (1.026 kWh/°F)** to both modes:

- Occupied: $\hat{y} = (\beta_0 + \beta_2) + 1.026 \cdot T = -39.0 + 1.026 \cdot T$
- Unoccupied: $\hat{y} = \beta_0 + 1.026 \cdot T = -56.9 + 1.026 \cdot T$

The occupancy premium is a **flat 17.85 kWh/hr** at every outdoor temperature — the same gap at a cold 40°F night as on a hot 105°F afternoon.

⚠ Is this physically realistic? On a cold night, does occupancy really add as much cooling demand as on a 105°F summer afternoon?

Categorical Predictors with More Than Two Levels

For a variable with k categories (e.g., Season has 4), Python creates $k - 1$ dummy variables. One category becomes the **reference level** (absorbed into the intercept).

Marking a predictor as **categorical** makes the software build the indicator columns for you — one per season, minus the one held back as the baseline.

The tutorial covers the notation.

The output shows coefficients for `C(Season)[T.Spring]`, `C(Season)[T.Summer]`, `C(Season)[T.Winter]` — each is the predicted extra kWh/hr for that season **vs. Fall**, holding outdoor temperature constant.

⚠ When `Outdoor_Temp_F` is already in the model, Season adds almost nothing (R^2 rises from 0.601 to 0.611 only). Outdoor temperature and season are measuring the same underlying phenomenon — how warm it is outside.

Part 3: Interaction Effects & Model Selection

When the slope itself depends on who's in the building

Predict — Same Slope or Different?

60 seconds · discuss with your neighbor

Model 2 says occupancy adds a **flat 17.9 kWh/hr** to cooling at any temperature.

But think physically: on a cold 40°F night, does occupancy add the same cooling demand as on a 105°F summer afternoon?

- **A)** The same slope for both — occupancy is a flat shift, temperature sensitivity is the same
- **B)** Different slopes — occupied buildings are more sensitive to outdoor temperature

✓ Different Slopes — The Interaction Model

Occupancy amplifies temperature sensitivity 2.5×

Mode	Slope (kWh/°F)	Intercept (kWh)	What's driving it
Unoccupied	0.71	−31.2	Envelope conduction only
Occupied	1.78	−78.3	Envelope + people + equipment + lighting

At 105°F:

- Unoccupied: $0.71 \times 105 - 31.2 = \mathbf{43.3 \text{ kWh/hr}}$
- Occupied: $1.78 \times 105 - 78.3 = \mathbf{108.6 \text{ kWh/hr}}$ — 2.5× higher

⚠ The additive Model 2 predicted a flat 17.9 kWh/hr occupancy premium — the interaction model shows it scales from near zero on cold nights to enormous on hot afternoons.

Interaction Terms — When the Slope Changes

The additive model assumes occupancy shifts cooling **up or down by a fixed amount**, regardless of temperature. But is that true?

Intuition: on a 40°F day, the building barely needs cooling whether occupied or not — the occupancy premium is small. On a 105°F day, a fully occupied building adds much more heat from people, computers, and lights on top of an already-hot exterior. The occupancy premium is larger.

This means the **slope itself depends on occupancy**:

- Unoccupied slope: **0.71 kWh/°F** (mostly envelope conduction — heat through walls and windows)
- Occupied slope: **1.78 kWh/°F** (envelope + internal loads from people, lighting, equipment)

The occupied building is **2.5× more sensitive** to outdoor temperature than the unoccupied one.

Interaction Model – Two Equations in One

The interaction model replaces one line with two:

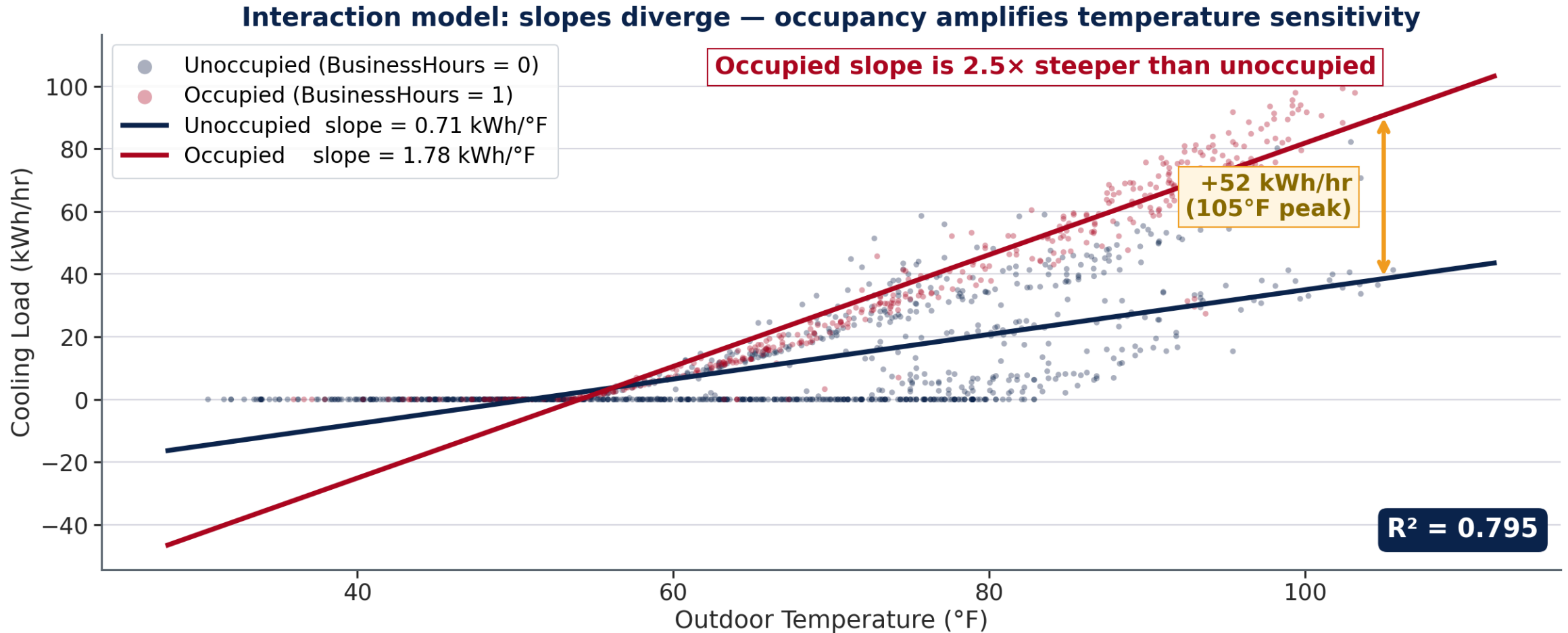
$$\hat{y} = \beta_0 + \beta_1 T + \beta_2 BH + \beta_3 (T \times BH)$$

The term $\beta_3 (T \times BH)$ is the **interaction term**: it adds extra slope when $BH = 1$.

Condition	Equation	Interpretation
Unoccupied ($BH = 0$)	$\hat{y} = \beta_0 + \beta_1 T$	baseline slope = 0.71 kWh/°F
Occupied ($BH = 1$)	$\hat{y} = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)T$	steeper slope = 1.78 kWh/°F

In Python: `smf.ols('Cooling_kWh ~ Outdoor_Temp_F * BH', data=df)` — the `*` automatically creates both main effects and the interaction term.

Interaction Model – Two Diverging Lines



Interaction Model — What the Gap Means

The occupied line is **2.50× steeper** than the unoccupied line. At 105°F the occupied building demands ~52 kWh/hr more than the unoccupied one — a gap that grows with temperature, not a flat shift. The additive model misses this entirely.

Which Model Wins?

60 seconds · pick A, B, or C

Three models, all using the same dataset. Two metrics reported:

Model	Predictors	k	R ²	AIC
A	Outdoor_Temp_F only	2	0.601	73,407
B	Outdoor_Temp_F + BH + Season	6	0.702	70,854
C	Outdoor_Temp_F × BH (interaction)	4	0.795	67,571

Which model would you select for an engineering report?

- **A)** Model A — simplest, fewest assumptions
- **B)** Model B — more predictors captured, higher R²
- **C)** Model C — best R² and lowest AIC with only 4 parameters

✓ Model C — Best on Both Metrics

Higher R^2 , lower AIC, and fewer parameters than Model B

Model	Predictors	k	R^2	AIC	Δ -AIC	Verdict
A	Temp only	2	0.601	73,407	+5,836	Baseline
B	Temp + BH + Season	6	0.702	70,854	+3,283	Season adds complexity without payoff
C	Temp $\times$ BH	4	0.795	67,571	0	✓ Wins on every criterion

Model B uses 6 coefficients for $R^2=0.702$. Model C uses 4 coefficients for $R^2=0.795$. The interaction term extracts more information than Season does — with fewer parameters.

Δ -AIC > 10 is decisive. Model C beats B by 3,283 — a commanding margin.

Model Comparison — R² Progression

Each model adds one idea. The R² gains tell you whether that idea was worth adding.

Model	Predictors	R ²	What it added
Week 9 simple	Outdoor_Temp_F	0.601	Temperature baseline
Model 2	+ BusinessHours	0.697	Occupancy: +9.6 pp
Model 3	+ Season	0.611	Season: only +1.0 pp (collinear)
Model 4	+ BH + Season	0.702	Season adds 0.5 pp on top of BH
Model 5	Temp × BH interaction	0.795	Slopes differ: +9.8 pp

The interaction model (Model 5) is clearly the best fit. But it comes at the cost of a more complex equation with four coefficients instead of three.

Model Selection — AIC

AIC (Akaike Information Criterion): lower = better. Every extra coefficient costs a penalty — even when R^2 increases.

Model	Predictors	k	R^2	AIC	Δ -AIC	Verdict
Model 1	Temp	2	0.601	73,407	+5,836	Baseline
Model 2	Temp + BH	3	0.697	70,992	+3,421	BH earns its place
Model 3	Temp + Season	5	0.611	73,181	+5,610	⚠️ 3 extra params, tiny R^2 gain
Model 4	Temp + BH + Season	6	0.702	70,854	+3,283	Season marginal on top of BH
Model 5	Temp × BH	4	0.795	67,571	0	✅ Best R^2 and lowest AIC

Δ -AIC > 10 is decisive. Model 5 beats Model 2 by 3,421 — not close. Model 3 (Season) uses 5 parameters for a 0.010 R^2 gain; its AIC barely improves vs. Model 1. The interaction term (Model 5) extracts 0.098 more R^2 than Model 2 using only one additional parameter.

Residuals — Did Adding Predictors Help?

The residual plot is the best diagnostic for model fit.

Week 9 simple model

- Clear fan shape: residuals spread wider at high temperatures
- Cluster of positive residuals at $T < 45^{\circ}\text{F}$ (chiller off, model predicts negative cooling)
- Pattern = the model is systematically wrong in two temperature ranges

Model 5 (interaction)

- Fan shape mostly gone — occupied and unoccupied hours now follow separate lines
- Low-temperature cluster reduced — separate intercepts let unoccupied fit lower
- Still some structure, but much closer to random scatter

The goal is residuals that look like **random noise** — no trend, no fan shape, no clusters. Perfect randomness means the model has captured all systematic variation.

Part 4: Connections

MLR across CE disciplines and into machine learning

MLR Across CE Domains

Multiple regression with categorical and interaction predictors spans every CE subfield:

Structural Engineering

- Predictors: dead load + live load + wind (each independent)
- Interaction: wind $\times$ height (taller buildings amplify more)

Water Resources

- Predictors: rainfall + soil moisture + season dummy
- Season dummy: wet/dry season shifts turbidity response

Transportation

- Speed $\sim$ volume + truck% + rain dummy (shifts speed-flow curve)
- VIF check: rush-hour dummy vs. volume often collinear

Energy Systems

- EUI $\sim$ HDD + CDD + occupancy type (k-level) + vintage dummy
- Vintage $\times$ CDD: older buildings react more sharply to heat

Python Quick Reference — Week 10

Concept	Python
Fit multiple regression	<code>smf.ols('y ~ x1 + x2', data=df).fit()</code>
Create dummy variable	<code>df['BH'] = (df['BH']=='Yes').astype(int)</code>
Auto-dummies (k levels)	<code>C(Season)</code> inside formula string
Interaction term	<code>smf.ols('y ~ x1 * x2', data=df).fit()</code>
Check VIF	<code>variance_inflation_factor(X.values, i)</code>
Compare models	<code>.aic</code> attribute on fitted model

Model Selection Checklist

Apply in order when building any multi-predictor model:

1. **Start** with physically motivated predictors
2. **Check VIF** — remove or combine if $VIF > 5$
3. **Add interaction** if physical reasoning supports different slopes by group
4. **Compare** Adj. R^2 and AIC — always plot residuals before reporting

Bridge to Machine Learning — Feature Engineering

The ideas from today are the foundation of how ML models work.

Statistics term	ML equivalent	Connection
Multiple regression	Linear model / ridge / lasso	Same equation, different solver
Adding predictors	Feature engineering	Selecting which x-variables to use
Dummy variable	One-hot encoding	Same concept, standard preprocessing
Interaction term ($T \times BH$)	Feature cross	Feature product used in boosting
Adjusted R^2	Validation score / RMSE	Both penalize overfitting

The **interaction model** is exactly how tree-based models (random forests, gradient boosting) work: each split says "if $BH=1$, use this slope; if $BH=0$, use that slope." Today's four-coefficient model is the simplest possible version of that idea.

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Nov 3
In-Class Exercise	The Colab notebook — all sections, plots, and written answers	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Nov 3; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 11: Hypothesis Testing

This week you use p-values to test whether each coefficient is real. Week 11 formalizes this idea from first principles.

The setup: occupied hours average 38.9 kWh/hr; unoccupied average 10.6 kWh/hr. Is the 28.3 kWh gap real, or could it be sampling noise?

Week 11 tools:

- **Two-sample t-test:** tests whether two group means differ
- **H_0 :** the difference is zero — the groups are the same
- **p-value:** probability of seeing this gap if H_0 were true
- **Connection:** the BH p-value in today's regression is doing this same test